1. Concept Document

Team Identification:

[redacted]

Title: Dual Protocol

Premise:
In an earth-like world, 2 individuals, an ordinary civilian and a mech operator, find themselves in the middle of an invasion, where their planet is being attacked.

Player Motivation:
Players try to discover what's going on and fight to protect their world. Uncover new gear, and abilities. Each mission pushes them closer to the core of the incident.

Unique Selling Proposition:

- Online 2-player cooperative campaign
- Titanfall-style parkour + mech combat, co-op attack combos
- Procedurally generated buildings & destructible environments

Target Market / Age: 16–45, action and co-op games fans, sci-fi and mech fans

Genre: Sci-fi Shooter / Action / Co-op

Expected HW Requirements:
At the moment, any laptop or desktop.

Competitive Analysis:

1. **Titanfall 2** - Parkour and mech combat inspiration
2. **Monster Hunter World** - Melee combat and Online Co-op
3. **Battlefield** - Procedural Building destruction
4. **It Takes Two** - Co-op driven story and gameplay
5. **Doom** - Level design/progression

Goals: Deliver a fast-paced, immersive, action packed co-op experience, showcasing procedural generation & real-time destruction and combining mech and humanoid gameplay with new combat abilities.

2. Game Proposal

Title, Hook, Goals: Dual Protocol is a co-op melee action game where a player controls a human character and another controls a mech, both with their unique abilities, in missions with procedural building generation/destruction.

Story Synopsis & Characters:

Backstory: A normal world is unexpectedly invaded by unknown entities, and only two people are capable and available to face them head-on.

Characters: An ordinary civilian (Player 1) living an average life in the city. A Mech Operator (Player 2), on duty. Entity's Commander, the main antagonist.

Gameplay and Main Mechanics:

- First or Third-person melee and ranged combat with wall-running, sliding, and double-jumps. Mech and guns included.
- Randomly generated building layouts in missions
- Buildings destructible based on structural logic
- Co-op gameplay with interactions/combos between characters
- Missions include sabotage, data retrieval, and boss fight

Technology / Libraries / Algorithms:

Plugins: Animator controllers, Cinemachine, NavMesh, *more to be determined*…

Custom made: Procedural generation of buildings with editing and procedural destruction given rules. Multiplayer Support (co-op).

Main Components Diagram:

Both players will have a player component, but dedicated components for their respective combat systems. The movement controller should be the same and might be limited/adjusted by other components, for unique input's and/or interactions. (Figure 1)

Art & Audio Features:

Stylized sci-fi low-poly environment (Figure 2), sound effects for footsteps and movement (including mech), weapons and abilities, destruction and background music.

Production Details:

Team:

[redacted] / [redacted] / [redacted]

Roles:

- [redacted]: Procedural Generation & Gameplay Mechanics
- [redacted]: Network & Combat Systems
- [redacted]: Art, UI/UX, Audio Design

Tasks:

- **Asset Search:** Find the best assets for the game (Everyone)
- **World Building:** Design and structure the game world (Everyone)
- **Character Design:** Create characters and visual styles (André Branco)
- **City Base Building:** Structure and plan city base (João Silveira)
- **Enemy Design:** Conceptualize enemy appearances (Ana Carolina)

- **Character's Movement:** Implement player and mech movement systems (João Silveira & André Branco)
- **Mech Movement:** Create control system for mech traversal (André Branco)
- **Enemy AI:** Program enemy behavior and pathfinding (Ana Carolina)
- **Procedural Building Generator:** Create ruleset system (João Silveira)

- **Combat Gameplay:** Handle player and enemy combat (Ana Carolina & André Branco)
- **Mech Combat:** Develop combat features for mechs (André Branco)
  Destruction Mechanics: Rule-based collapse (João Silveira)
- **UI & Menus:** Design game interface and navigation (Ana Carolina & André Branco)
- **Network Integration:** Handle co-op logic (André Branco & Ana Carolina)

- **Bug Fixing:** Identify and resolve bugs (Everyone)
- **Combined Animations:** Sync and refine movement/attack animations (André Branco & Ana Carolina)
- **Sound Effects:** Create and integrate game audio (Ana Carolina)
- **Final Report:** Compile documentation and project summary (Everyone)

Gantt Chart: *(Include with timeline: April 30 – June 3)* (Figure 3)

Risk Analysis:

- **Procedural destruction** may require complex structural rules or physics

- **Co-op issues,** like desync or disconnection.

- **Time constraints** for art & combat design and animations.

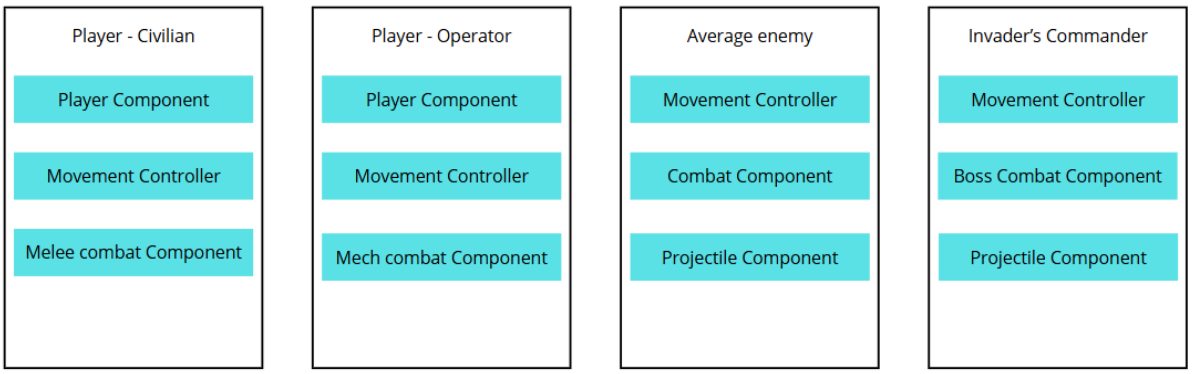

Figure 1: Component Diagram

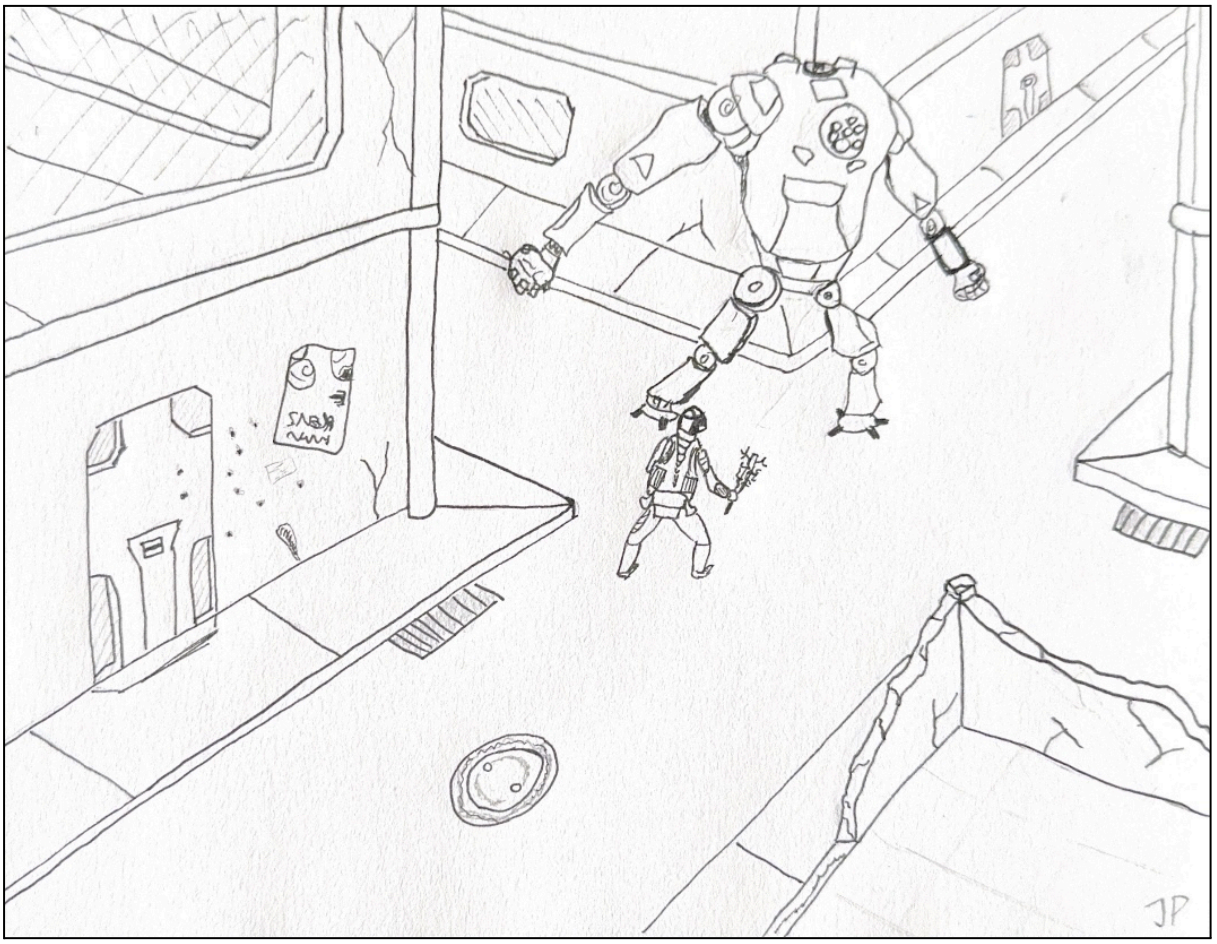

Figure 2: Concept Art

TASK	ASSIGNED TO	START	END
Phase 1 - Begining			
Find assets	Everyone	07/05/25	11/05/25
World Design	Everyone	11/05/25	15/05/25
Character Design	André Branco	13/05/25	18/05/25
City Base Building	João Silveira	13/05/25	16/05/25
Enemy design	Ana Carolina	13/05/25	18/05/25
Phase 2 - Ground Work			
Human movement	João Silveira & André Branco	16/05/25	20/05/25
Mech movement	André Branco	18/05/25	22/05/25
Enemy AI	Ana Carolina	18/05/25	23/05/25
Building Generation	João Silveira	19/05/25	23/05/25
Phase 3 - Main Features			
Human combat	Ana Carolina & André Branco	20/05/25	23/05/25
Mech Combat	André Branco	22/05/25	25/05/25
Building Destruction	João Silveira	23/05/25	30/05/25
UI & Menus	Ana Carolina & André Branco	24/05/25	28/05/25
Networking	André Branco & Ana Carolina	25/05/25	30/05/25
Phase 4 - Testing and Polishing			
Bug Fixing	Everyone	27/05/25	03/06/25
Combined animations	André Branco & Ana Carolina	27/05/25	31/05/25
Sound Effects	Everyone	27/05/25	30/05/25
Final Report	Everyone	27/05/25	03/06/25

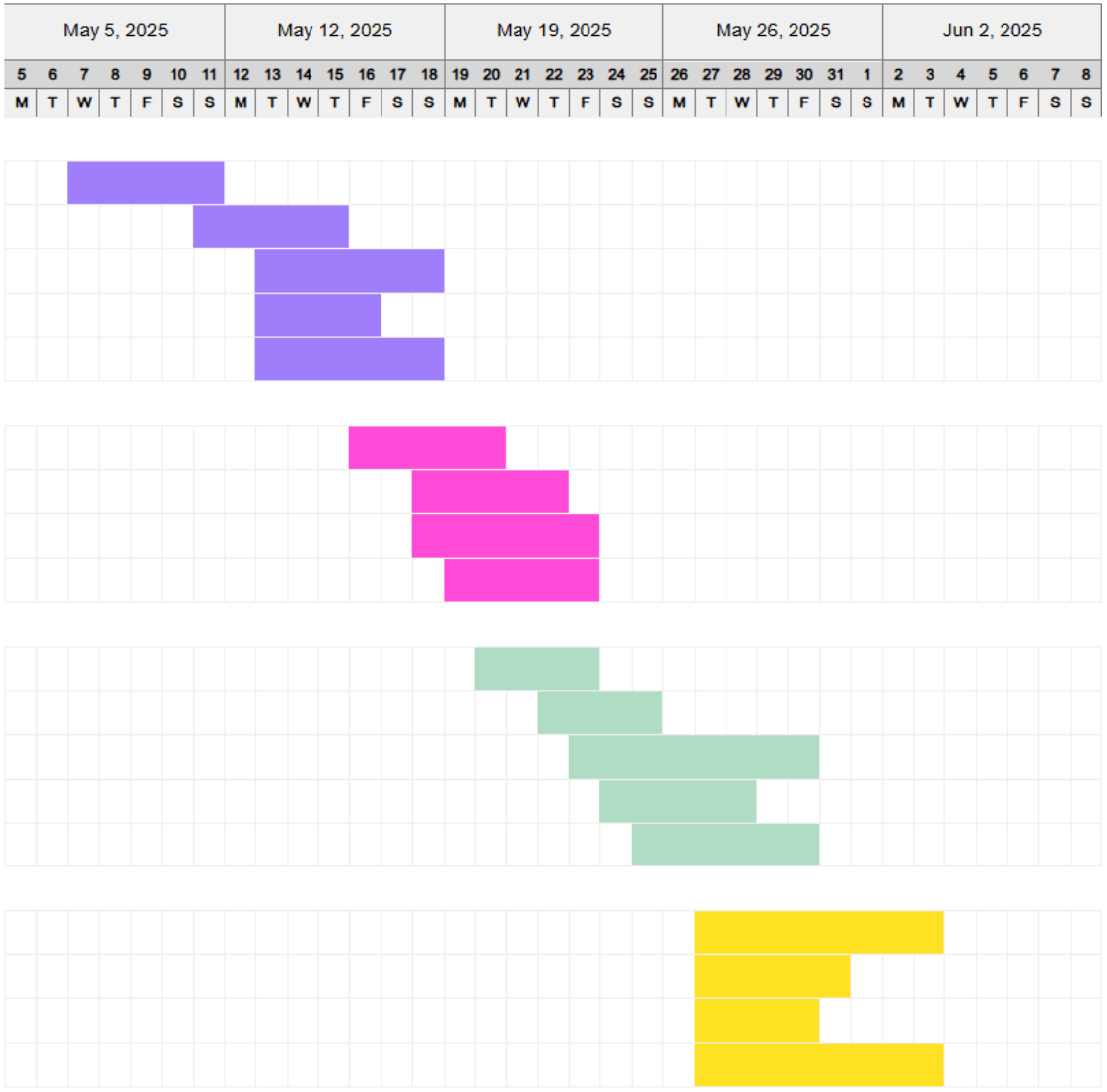

Figure 3: Gantt Chart